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Paleomagnetic and geochronological study of Carboniferous forearc basin rocks in the Southern New England Orogen (Eastern Australia)

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ABSTRACT

We present results of a paleomagnetic study from Carboniferous forearc basin rocks that occur at both limbs of the Texas Orocline (New England Orogen, eastern Australia). Using thermal and alternating field demagnetizations, two remanence components have been isolated from rocks sampled from the Emu Creek terrane, in the eastern limb of the orocline. A middle-temperature Component M is post-folding and was likely acquired during low-temperature oxidation at 65–35 Ma. A high-temperature Component H is pre-folding, but its comparison with the paleomagnetic data from coeval rocks in the northern Tamworth terrane on the other limb of Texas Orocline does not indicate rotations around a vertical axis, as expected from geological data. A likely explanation for this apparent discrepancy is that Component H postdates the oroclinal bending, but predates folding in late stages of the 265–230 Ma Hunter Bowen Orogeny. The post-Kiaman age of Component H is supported by the presence of an alternating paleomagnetic polarity in the studied rocks. A paleomagnetic study of volcanic and volcaniclastic rocks in the Boomi Creek area (northern Tamworth terrane) revealed a stable high-temperature pre-folding characteristic remanence, which is dated to c. 318 Ma using U–Pb zircon geochronology. The new paleopole (37.8°S, 182.7°E, A_{95} = 16.2°) is consistent with previously published poles from coeval rocks from the northern Tamworth terrane. The combination of our new paleomagnetic and geochronological data with previously published results allows us to develop a revised kinematic model of the New England Orogen from 340 Ma to 270 Ma, which compared to the previous model, incorporates a different orientation of the northern Tamworth terrane at 340 Ma.

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1. Introduction

The New England Orogen (NEO), which is the youngest segment of the Tasmanides in eastern Australian (Fig. 1), developed as an accretionary orogen during the late Paleozoic and early Mesozoic along the Pacific margin of the Gondwana supercontinent (Cawood, 2005). The orogen predominantly consists of Devonian–Carboniferous subduction related units (magmatic arc, forearc basin and accretionary complex), which are partly covered or intruded by younger sedimentary and magmatic rocks (Fig. 1b). The southern part of the NEO exhibits a doubly vergent oroclinal structure with the southern (Manning – Nambucca oroclines) and northern (Texas – Coffs Harbour oroclines) segments displaying S- and Z-shaped geometries, respectively. It is generally accepted that the oroclines formed during the Early Permian (299–270 Ma), but their geodynamic evolution is controversial (e.g., Murray et al., 1987; Korsch and Harrington, 1987; Glen, 2005; Offler and Foster, 2008; Cawood et al., 2011; Glen and Roberts, 2012; Rosenbaum, 2012a; Rosenbaum et al., 2012; Shaanan et al., 2015), particularly due to the scarcity of robust constraints on the kinematics of oroclinal bending.

The largest and most prominent curvature in NEO is the Texas Orocline (Fig. 1b). Part of the orocline is concealed beneath younger sedimentary rocks of the Surat Basin (Brooke-Barnett and Rosenbaum, 2015), but the curvature is clearly evident in aeromagnetic images (e.g., Brooke-Barnett and Rosenbaum, 2015), magnetic fabric (Aubourg et al., 2004; Mochales et al., 2014), and in the curved structural fabric of accretionary complex rocks (Lennox and Flood, 1997; Li et al., 2012). An additional independent marker of the orogenic curvature is the occurrence of Carboniferous forearc basin

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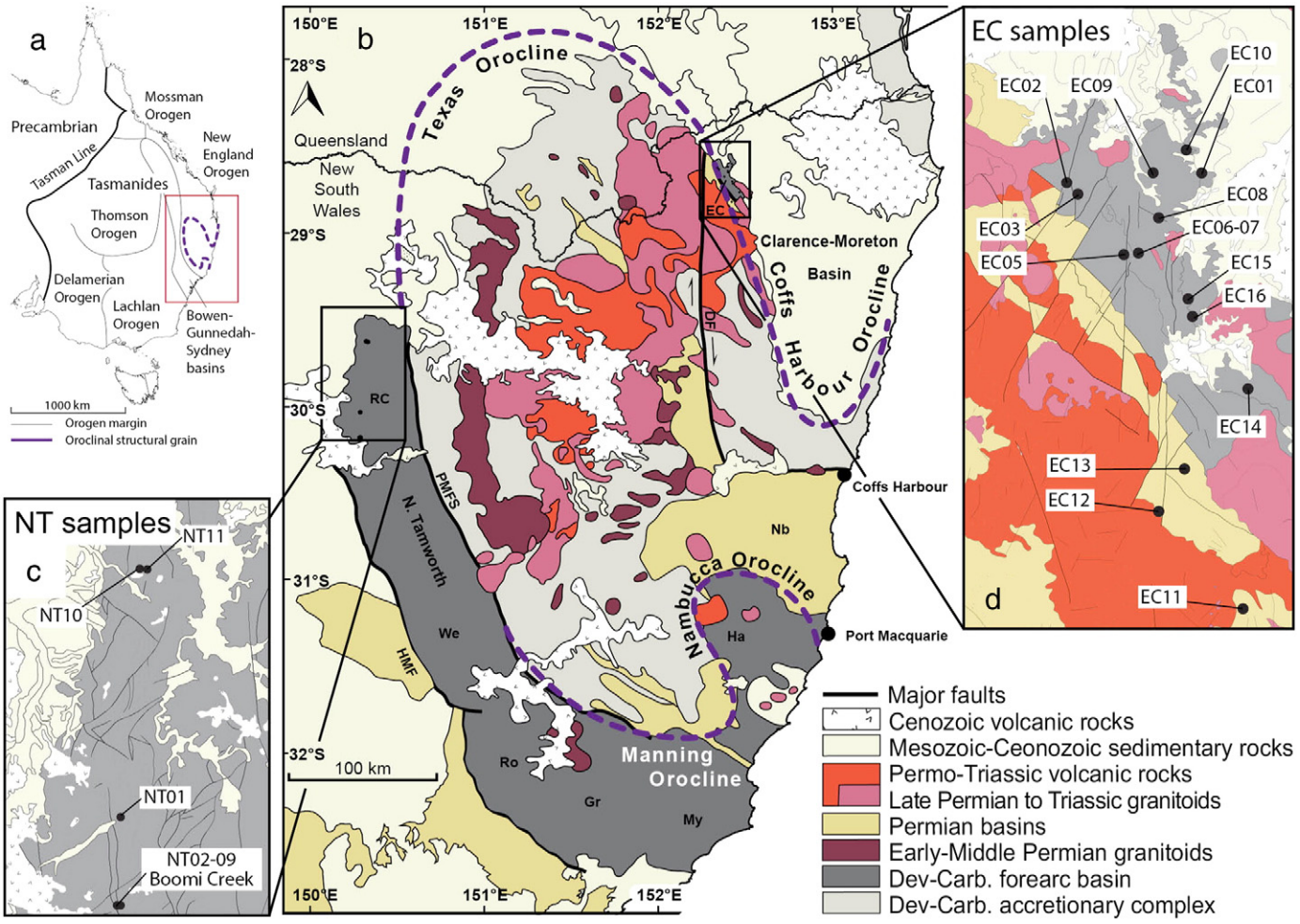


Fig. 1. (a) Location of the New England Orogen in the easternmost Tasmanides; (b) simplified tectonostratigraphic map of the southern New England Orogen. DF, Demon Fault; Ec, Emu Creek; Gr, Gresford; Ha, Hastings; Mb, Mount Barney; My, Myall; Na, Nandewar; Nb, Nambucca; PMFS, Peel Manning Fault System; HMF, Hunter–Mooki Fault; Rc, Rocky Creek; Ro, Rouchel; We, Werrie. (c) Geological map and sample locations in the northern Tamworth terrane. (d) Geological map and sample locations in the Emu Creek terrane.

rocks that are suggested to be correlative on both limbs of the Texas Orocline (Hoy et al., 2014), with the Emu Creek terrane exposed on the eastern limb, and the Rocky Creek Block (northernmost part of the Tamworth terrane) exposed on the western limb (Fig. 1b). The correlation between the Emu Creek and northern Tamworth terranes (Fig. 2) was based on similar biostratigraphic faunal assemblages, depositional environment, provenance, and geochronology (Hoy et al., 2014), but their spatial relations prior to oroclinal bending have not been established.

Paleomagnetic studies provide a powerful tool to reconstruct rotations of continents and terranes in general, and are thus of a particular importance for oroclinal structures (e.g., Johnston, 2001; Van der Voo, 2004; Weil et al., 2010; Gutiérrez-Alonso et al., 2012). In the southern NEO, Schmidt et al. (1994) studied the Kullatine Formation in the Hastings Block (Fig. 1b) and suggested 130° clockwise or 230° anticlockwise post-Serpukhovian rotation of this block with respect to the Australian craton. Paleomagnetic data from Visean rocks in the Rouchel, Gresford and Myall terranes (Fig. 1b) of the Manning Orocline (Geeve et al., 2002) were interpreted as post-Visean anticlockwise block rotations of 80°, 80° and 120°, respectively. These rotations are consistent with the hypothesised geometry of the southern segment of the oroclinal structure (e.g., Rosenbaum, 2012b; Li and Rosenbaum, 2014). However, Cawood et al. (2011) have proposed a different kinematic model and demonstrated that such large apparent rotations could partly be explained by long distance movements of these blocks in medium or high latitudes in a way suggested by Cox (1980) and Debiche et al. (1987) for terranes in western North America.

The aim of this paper is to refine previous kinematic reconstructions of the southern NEO (Cawood et al., 2011) using new paleomagnetic and geochronological data. We attempt to quantify vertical axis block rotations around the Texas Orocline, and to test whether such rotations are consistent with suggested models for oroclinal bending (e.g. Murray et al., 1987; Korsh and Harrington, 1987; Glen, 2005; Offler and Foster, 2008; Cawood et al., 2011; Glen and Roberts, 2012; Rosenbaum, 2012a; Rosenbaum et al., 2012; Shaanan et al., 2014). We present new paleomagnetic data from Carboniferous rocks in the Emu Creek and northern Tamworth terranes (Fig. 1), and U–Pb geochronological results that further refine stratigraphic relations in the northern Tamworth terrane.

2. Geology and sampling

The northern Tamworth terrane is commonly subdivided into two blocks, the Rocky Creek Block in the north and the Werrie Block in the south (Figs. 1 and 2). These blocks consist of predominantly Devonian and Carboniferous volcanic, volcaniclastic and sedimentary rocks, which were deposited in a forearc basin setting (Day et al., 1978). Previous paleomagnetic studies have been conducted on Visean to Kasimovian rocks from both the Rocky Creek and Werrie blocks

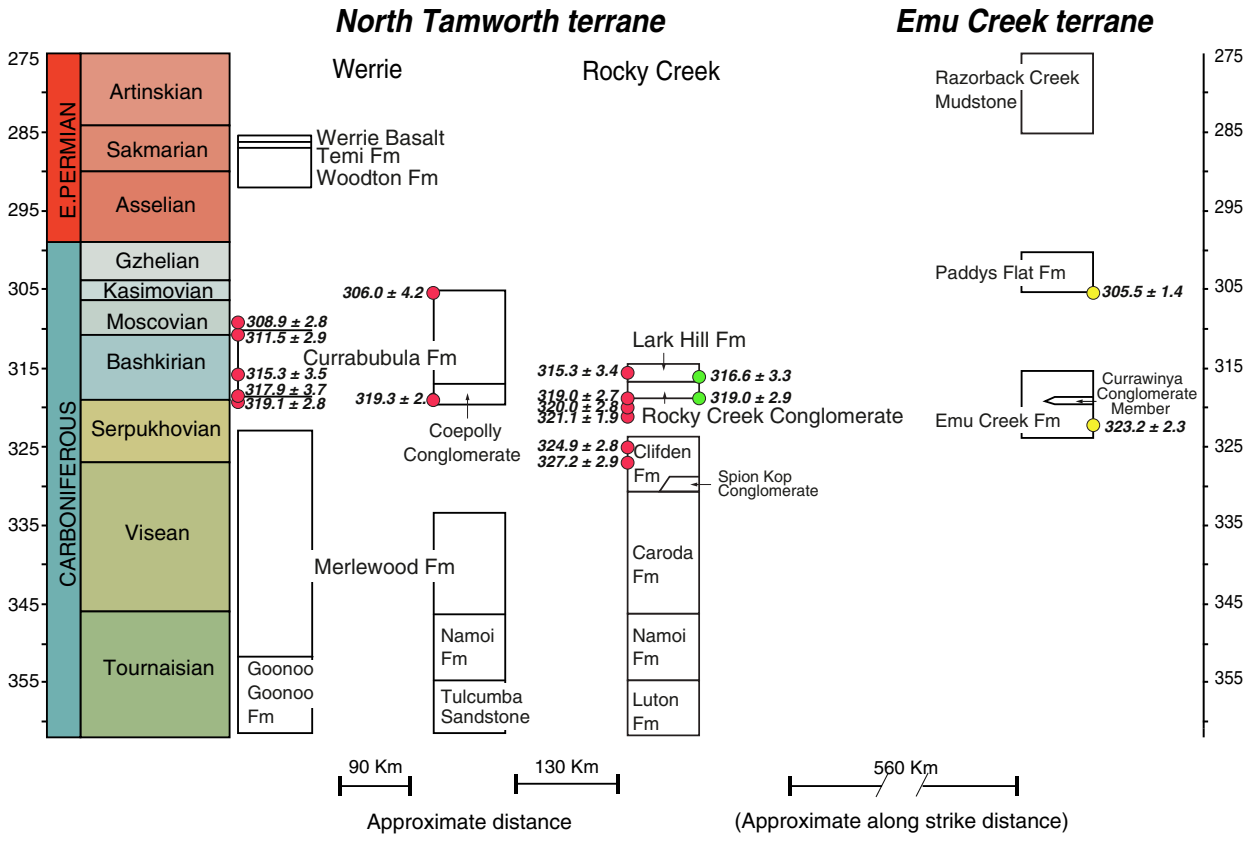


Fig. 2. Chronostratigraphic correlation of the sampling sits. Red circles denote SHRIMP zircon analyses with usage of AS3 standard and in one case (Roberts et al., 2003; Roberts et al., 2006; Roberts and James, 2010). LA-ICP-MS zircon analyses (TEM and GJ1 standards) of igneous rocks (this study) are denoted by green circles. LA-ICP-MS detrital zircon data (Hoy et al. 2014) marks maximum depositional age constraints and are denoted by yellow circles.

(Opdyke et al., 2000; Klootwijk, 2002, 2003). In addition, stratigraphic and U–Pb SHRIMP geochronological investigations (Roberts et al., 2003; Roberts et al., 2006; Roberts and James, 2010) have significantly improved the resolution of chronostratigraphic relationships in the Rocky Creek and Werrie blocks (Fig. 2).

Rocks in the northern Tamworth terrane are part of a fold-thrust belt, and have been moderately deformed and metamorphosed under low-grade (<200 °C) conditions (Woodward, 1995, Offler et al., 1997). Deformation is characterized by west vergent faults and non-cylindrical folds, which are oriented predominantly NNW–SSE, parallel to the structural grain of the orogen (Korsch, 1977). The eastern and western boundaries of the terrane are defined by major fault contacts, which are the Peel–Manning Fault System and the Hunter–Mooki Fault, respectively (Fig. 1).

The Emu Creek terrane is a fault-bounded block of ~45 × 20 km aligned along the eastern limb of the Texas Orocline (Fig. 1b, d). The block consists of slightly deformed, immature, shallow-marine sedimentary rocks of the Serpukhovian to Gzelian Emu Creek and Paddys Flat formations (Fig. 2; Hoy et al., 2014). The age of the Emu Creek Formation is constrained by the occurrence of fossils from the *Levipustula levis* assemblage (Voisey, 1936; McCarthy et al., 1974; Roberts et al., 1995), which indicates Bashkirian–Serpukhovian (323–315 Ma) age. Recent detrital zircon study of this terrane yielded maximum depositional age constraints for the Emu Creek Formation and unconformably overlying Paddys Flat Formation of c. 323 Ma and c. 306 Ma, respectively (Hoy et al., 2014). The age constraints, in conjunction with paleontological data, suggest that the Emu Creek and Paddys Flat formations can be considered as a broad correlative to the Currabubula Formation in the northern Tamworth terrane (Fig. 2; Hoy et al. 2014).

The Emu Creek terrane has been affected by gentle to tight NE-striking folds and both normal and strike-slip faults (Hoy et al. 2014), although the timing of deformation is not quite clear. Contractional deformation may have occurred at 265–230 Ma during the so-called Hunter–Bowen Orogeny (e.g., Holcombe et al., 1997).

We collected 106 oriented samples from 14 sites in the Emu Creek terrane (Emu Creek and Paddys Flat formations; Fig. 1d), for paleomagnetic purposes. Both magnetic and sun compasses have been used for sample orientation and 1 to 3 specimens have been obtained from each drill sample. In addition, 99 oriented samples were collected from 11 sites in the Rocky Creek Block (Fig. 1c). Most sampling locations (8 out of 11 sites) are from volcanic and volcaniclastic rocks exposed in Boomi Creek (Fig. 1c). These rocks are mapped as part of the Visean Caroda Formation (Fig. 2; Brown et al., 1992), but Klootwijk (2003) suggested that their age is younger. To address this problem, we conducted U–Pb zircon geochronology on two samples from the Boomi Creek section (sites NT02 and NT04, see section 4). These two samples are separated by ~20 m of stratigraphic thickness of intermediate–felsic volcanic rocks that are continuously exposed along Boomi Creek.

3. Paleomagnetism

The paleomagnetic study has been carried out in a magnetically shielded room hosted in the paleomagnetic laboratory of the Istituto

Nazionale di Geofisica e Volcanologia (INGV) in Rome, Italy. Remanence composition was determined by detailed stepwise thermal demagnetization (≤20 steps, to 650 °C) using a 2G Enterprises DC-superconducting quantum interference device cryogenic magnetometer and a Pyrox shielded oven, and stepwise alternating field (AF) demagnetization yielded by three coils online with the magnetometer (≤10 steps, up to 100 mT). Magnetic susceptibility has been measured with KLY-2 kappa-bridge (Geofyzika, Brno).

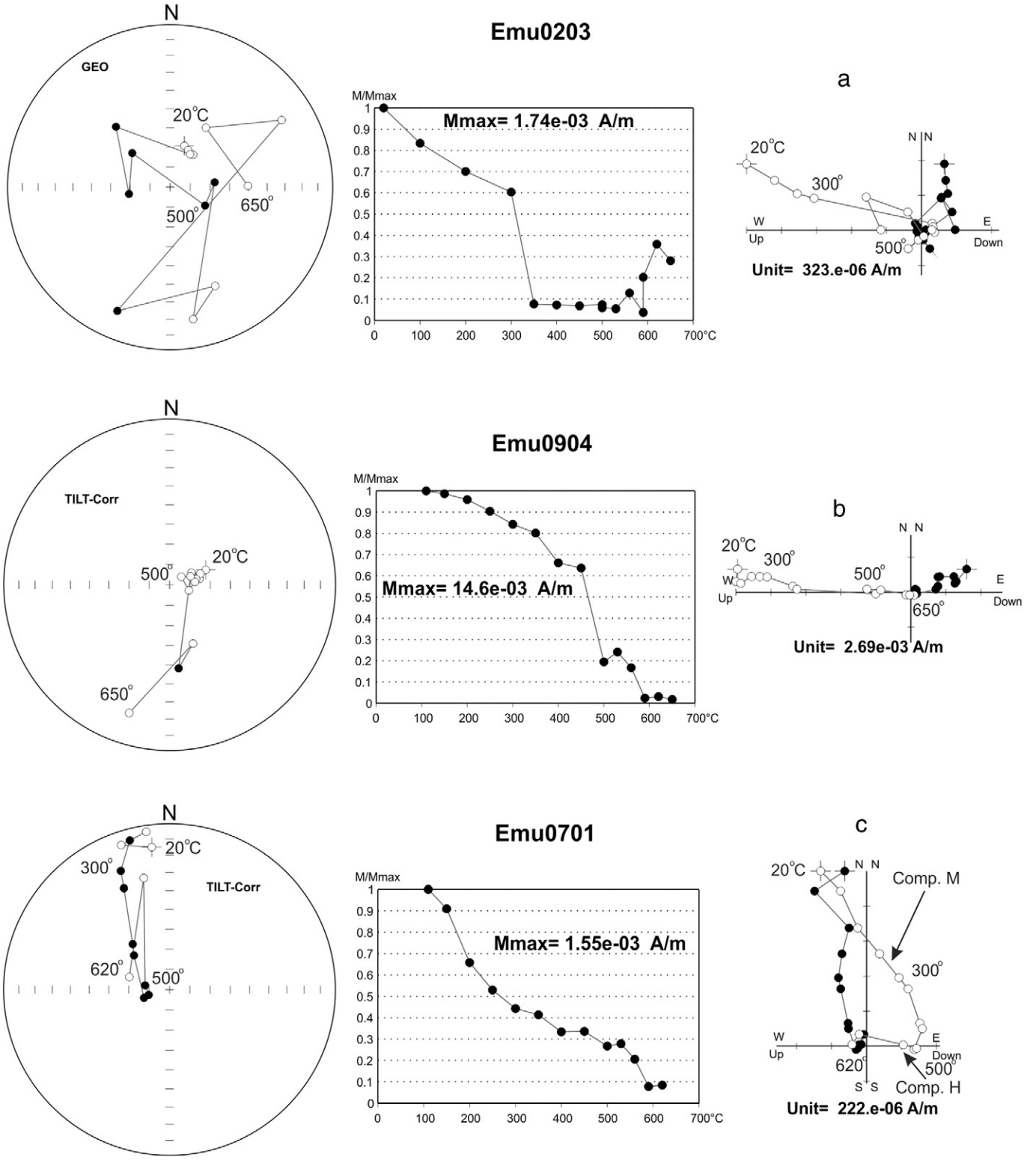


Fig. 3. Examples of thermal demagnetizations of sedimentary rocks of the Emu Creek and Paddys Flat formation. In orthogonal plots, open (closed) symbols show magnetisation vector endpoints in the vertical (horizontal) plane; curves show changes in intensity during demagnetisation. Stereoplots (Lambert projection) show upwards (downwards) pointing palaeomagnetic directions with open (closed) symbols; (a) M component; (b) H component; (c) both components.

Table 1
Paleomagnetic directions: Emu Creek and Paddys Flat formations (M component).

#	Site	N/n	Slat (°S)	Slong (°E)	Decl. (in situ) (°)	Incl. (in situ) (°)	k	α_{95} (°)	Decl. (tilt corrected) (°)	Incl. (tilt corrected) (°)	k	α_{95} (°)
1	Emu01 (PF)	11/7	28°37′09.35″	152°24′57.65″	9.2	−69.5	84.0	6.6	4.4	−54.7	84.0	6.6
2	Emu02 (EC)	6/5	28°38′30.52″	152°20′04.80″	6.0	−63.1	14.4	20.9	30.3	−43.5	14.4	20.9
3	Emu03 (EC)	7/6	28°38′59.37″	152°20′30.63″	48.7	−72.8	24.7	13.8	112.8	−70.5	24.7	13.8
4	Emu04 (EC)	11/9	28°41′25.72″	152°22′27.41″	341.0	−74.1	4.2	28.8	355.9	−49.1	4.2	28.8
5	Emu05 (PF)	7/2	28°41′20.49″	152°23′02.44″	228.0	76.4	–	–	107.4	73.8	–	–
6	Emu06 (PF)	5/4	28°39′57.14″	152°23′51.31″	258.0	69.6	27.0	18.0	191.1	66.7	27.0	18.0
7	Emu10 (EC)	5/1	28°32′55.84″	152°24′11.78″	4.2	−56.7	–	–	0.5	−66.5	–	–
8	Emu14 (EC)	8/5	28°46′57.01″	152°27′30.55″	341.9	−28.1	26.8	15.0	305.1	−26.0	34.4	13.2
9	Emu15 (PF)	8/2	28°43′15.20″	152°25′03.82″	357.5	81.8	–	–	64.6	68.5	–	–
10	Emu1601 (PF)	11/7	28°43′59.25″	152°25′13.09″	357.9	−50.1	16.3	15.4	356.7	−32.2	16.3	15.4
Mean (10 sites):					**8.5**	**−68.8**	**13.9**	**13.4**	**350.6**	**−64.0**	**7.4**	**19.0**
Paleomagnetic pole (in situ):					**61.1°S, 137.7°E,**			**A_{95} = 19.5°**				

N/N = number of demagnetised/used samples; Slat, Slong = locality latitude, longitude; Decl, Incl = site mean declination, inclination; k = best estimate of the precision parameter of Fisher (1953); α_{95} = the semi-angle of the 95% cone of confidence; PF = Paddys Flat Formation; EC = Emu Creek Formation.

3.1. Emu Creek terrane

The natural remanent magnetization (NRM) of the studied rocks ranges from 0.2 to 700 mA/m, and their magnetic susceptibility ranges from 8 to 14 × 10^{-5} SI units. The AF demagnetization in most cases was not effective and yielded inconsistent paleomagnetic directions. Therefore, the majority of specimens have been thermally demagnetized.

About 30% of the specimens do not carry stable remanence and show chaotic directional behaviour. Fifty seven specimens from 11 sites carry a stable low-middle temperature remanence component, which completely disappears above 300 °C (Fig. 3a). The further heating in most cases does not reveal any stable remanence. We suggest that this mid-temperature component (Component M) is carried by maghemite, which is the product of low-temperature oxidation or weathering (Dunlop and Özdemir, 1997; page 58). The sharp drop in the remanence intensity after heating over 300 °C (Fig. 3a) likely reflects the inversion of maghemite into hematite (Dunlop and Özdemir, 1997; page 59). Table 1 and Fig. 4 show the site-mean directions of Component M in situ and after the tilt correction. The component is bipolar. The reversal test of McFadden and McElhinny (1990) is indeterminate, and the angle between the means of two polarity sets is 149.7°. The fold test of

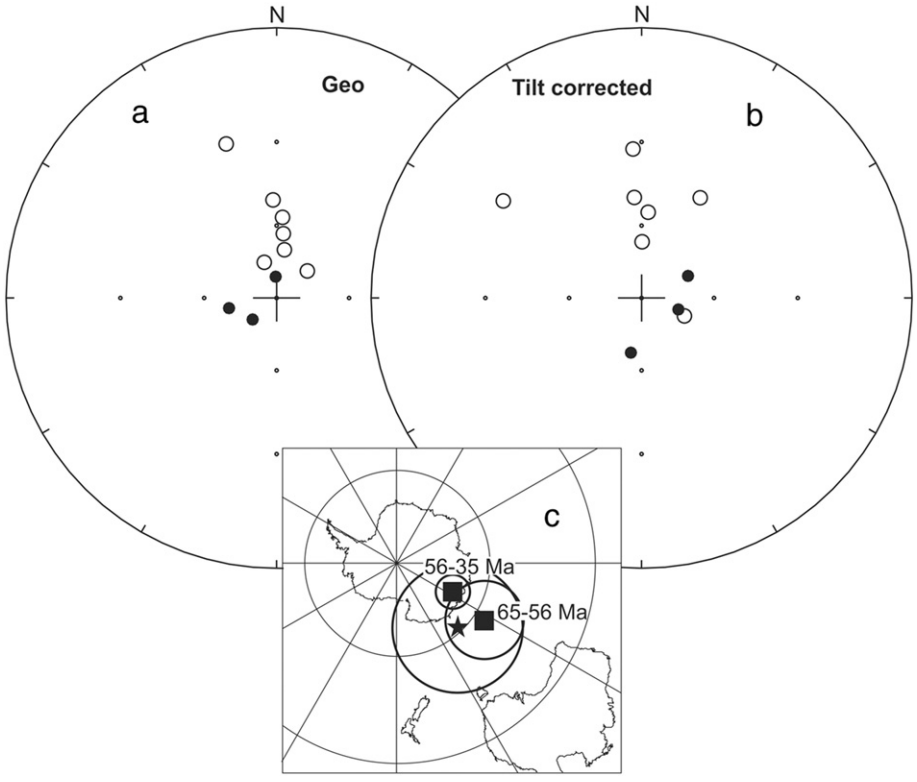


Fig. 4. (a–b) Stereoplots (stereographic projection) of the M component, in situ and tilt corrected correspondingly; (c) squares—Australian paleopoles (Table 6.14 of McElhinny and McFadden, 2000); star—paleopole calculated for the M component (Table 1).

Table 2
Paleomagnetic directions: Emu Creek and Paddys Flat formations (H component).

#	Sample	Slat (°S)	Slong (°E)	Decl. (in situ) (°)	Incl. (°)	MAD (°)	Decl. (tilt corrected) (°)	Incl. (°)
1	Emu0104 (PF)	28°37′09.35″	152°24′57.65″	279.4	64.2	9.3	248.3	68.3
2	Emu0206 (EC)	28°38′30.52″	152°20′04.80″	253.9	−84.8	6.4	57.8	−69.0
3	Emu0407 (EC)	28°41′25.72″	152°22′27.41″	205.6	−75.8	7.4	345.2	−76.6
4	Emu0701 (PF)	28°39′55.06″	152°23′49.05″	333.0	61.3	10.4	253.8	79.4
5	Emu0702 (PF)	“	“	298.8	66.8	9.6	227.2	65.5
6	Emu0705 (PF)	“	“	9.5	58.2	10.7	72.5	82.3
7	Emu0706 (PF)	“	“	336.5	73.2	11.9	197.1	75.9
8	Emu0707 (PF)	“	“	340.3	69.4	4.9	204.9	79.6
9	Emu0804 (PF)	28°38′06.28″	152°25′37.00″	42.9	−61.3	26.2	28.1	−51.2
10	Emu0901 (PF)	28°38′03.92″	152°23′37.67″	39.5	−73.9	4.0	13.8	−66.0
11	Emu0902 (PF)	“	“	4.0	−73.8	6.2	352.5	−62.0
12	Emu0903 (PF)	“	“	19.7	−76.8	7.2	359.5	−66.6
13	Emu0904 (PF)	“	“	110.3	−71.2	2.8	71.4	−76.6
14	Emu1001 (EC)	28°32′55.84″	152°24′11.78″	70.5	−54.8	12.8	84.2	−59.0
15	Emu1003 (EC)	28°32′55.84″	152°24′11.78″	195.1	−83.5	2.1	194.4	−73.5
16	Emu1601 (PF)	28°43′59.25″	152°25′13.09″	21.3	−86.3	10.3	357.8	−68.7
Mean (16 samples):								
In situ:	***D = 111.9°, I = −80.2°, k = 17.6, α_{95} = 9.0°, Plat = 21.0°S, Plong = 135.0°E, A_{95} = 16.0°***							
Tilt. corr.:	***D = 32.5°, I = −75.5°, k = 23.6, α_{95} = 7.6°, Plat = 46.2°S, Plong = 132.2°E, A_{95} = 14.7°***							

MAD = maximum angular deviation; Plat, Plong = latitude, longitude of the paleopole. PF = Paddys Flat Formation; EC = Emu Creek Formation. See also footnotes to Table 1.

McFadden (1990) is negative at the 99% confidence level. A fold test statistics SCOS (summation of cosine angles) is 1.072 in situ and 7.988 after tilt correction with critical value of 5.378. The Fisher's precision parameter is 13.9 in situ and 7.4 after the tilt correction. We conclude that Component M is post-folding. The mean in situ paleomagnetic direction for 10 sites is: D = 8.5°, I = −68.8°, k = 13.9, α_{95} = 13.4°. The corresponding paleomagnetic pole is at 61.1°S and 137.7°E (A_{95} = 19.5°). The circle of confidence of this pole encloses Australian paleopoles for Eocene and Paleocene (Fig. 4c; McElhinny and McFadden, 2000, pages 276–277). Hence, Component M could have been acquired during low-temperature oxidation, likely at 65–35 Ma. The presence of two polarities (Fig. 4a, b) suggests that acquisition of the M component possibly lasts one or more million years accommodating at least one geomagnetic reversal (e.g., McElhinny and McFadden, 2000).

Sixteen samples from 8 sites (5 sites from Paddys Flat Formation and 3 sites from Emu Creek Formation) carry a high-temperature (500–590 °C) Component H (Fig. 3b) and in few samples M and H components co-exist (Fig. 3c). The H component is also bipolar for both formations (Table 2; Fig. 5). As the directions are statistically undistinguishable between the Paddys Flat and Emu Creek formations, and considering their time constraints are ~17 m.y. apart, we calculated one combined mean direction and paleopole for the Emu Creek Block (Table 2). The reversal test of McFadden and McElhinny (1990) is positive with classification C, the angle between the means of two polarity sets is 173.3°. The fold test of McFadden (1990) is positive at the 99% confidence level, with a fold test statistics SCOS of 8.443 in situ and 0.172 after tilt correction with critical value of 6.516. The Fisher's precision parameter is 17.6 in situ and 23.6 after the tilt correction. We conclude that Component H is pre-folding. The mean tilt corrected

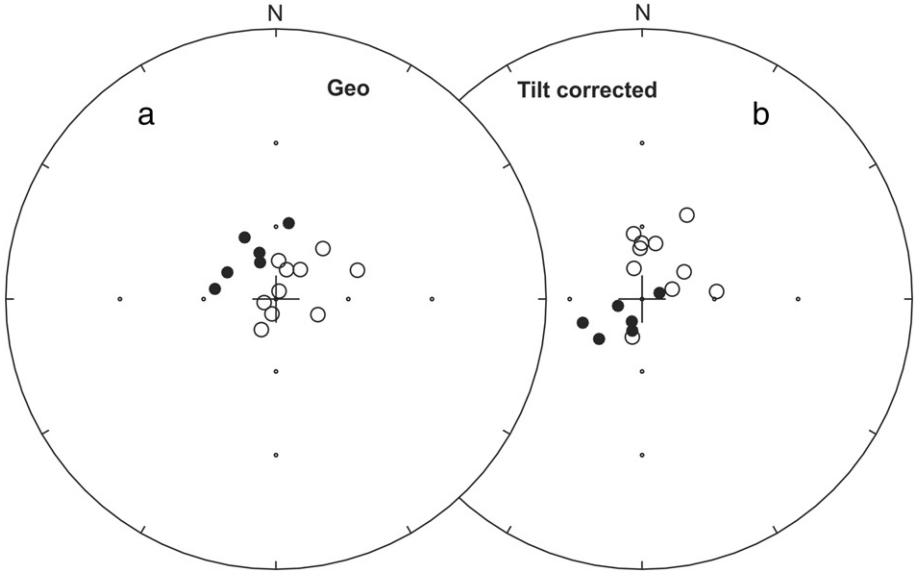


Fig. 5. Stereoplots of the H component in the Paddys Flat and Emu Creek formations (Table 2). (a) in situ and (b) tilt corrected.

Table 3
Paleomagnetic directions: North Tamworth terrane.

#	Site	N/n	Slat (°S)	Slong (°E)	Decl. (in situ) (°)	Incl. (in situ) (°)	k	α_{95} (°)	Decl. (tilt corrected) (°)	Incl. (tilt corrected) (°)	k	α_{95} (°)
1	NT02	5/5	30°11′41.10″	150°16′39.05″	88.3	52.4	147.6	6.3	106.4	70.1	147.9	6.3
2	NT03	5/5	30°11′41.23″	150°16′41.49″	77.9	49.1	45.5	11.5	94.1	60.6	45.5	11.5
3	NT04	5/5	30°11′41.22″	150°16′41.94″	89.5	59.3	28.6	14.6	118.7	76.0	28.6	14.5
4	NT05	3/3	30°11′41.00″	150°16′42.67″	93.8	55.8	68.4	15.0	134.1	63.0	69.0	15.0
5	NT06	5/2	30°11′40.96″	150°16′45.19″	104.9	66.8	–	–	158.3	75.0	–	–
6	NT07	5/3	30°11′40.43″	150°16′46.31″	96.9	52.4	151.9	10.0	124.6	62.4	149.3	10.1
7	NT08	5/3	30°11′38.75″	150°16′34.10″	95.8	37.3	78.4	14.0	98.4	71.3	78.4	14.0
8	NT10	6/5	29°38′22.33″	150°18′42.63″	277.9	28.2	27.3	14.9	4.4	75.4	27.2	14.9
9	NT11	6/3	29°38′32.69″	150°19′15.66″	106.0	58.0	248.6	7.8	196.5	82.9	247.5	7.9
	Mean (9 sites)				92.4	61.5	6.8	21.3	114.7	74.4	32.7	9.2
	Paleomagnetic pole (tilt corrected):				37.8°S, 182.7°E,			A_{95} = 16.2°				

See footnotes to Table 1.

paleomagnetic direction for 16 samples is: D = 32.5°, I = −75.5°, k = 23.6, α_{95} = 7.6°. The corresponding paleomagnetic pole is at 46.2°S and 132.2°E (A_{95} = 14.7°).

3.2. Northern Tamworth

The natural remanent magnetization (NRM) of the studied rocks ranges from 0.3 to 140.0 mA/m, and their magnetic susceptibility from about 10 to 100 × 10^{-5} SI units. Both thermal and AF demagnetizations have been applied.

A stable uni-polar Characteristic Remanent Magnetization (ChRM) with unblocking temperatures between 500 and 590 °C has been isolated in most of the studied samples (Table 3; Fig. 6). Thermal demagnetization suggests that magnetite is a dominant remanence carrier, but in some cases the presence of maghemite is possible (Fig. 6b). The maghemite dominated specimens from site NT09 have been excluded from the analysis.

The fold test of McFadden (1990) is positive at the 99% confidence level (Fig. 7). A fold test statistics SCOS is 5.473 in situ and 0.315 after tilt correction with the 99% critical value of 4.849. The Fisher's precision parameter is 6.8 in situ and 32.7 after the tilt correction. We conclude that the steep downward east ChRM is pre-folding. The corresponding paleomagnetic pole is 37.8°S, 182.7°E (A_{95} = 16.2°) (Table 3).

4. U–Pb geochronology

4.1. Sample preparation and analytical methods

To determine the age of the sequence that is exposed at Boomi Creek, we have conducted U–Pb geochronology on zircons using laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) at the Central Analytical Facility, Queensland University of Technology. Zircon crystals were separated from two samples of intermediate volcanic rocks from the Rocky Creek block (NT02 and NT04). Samples were crushed to <600 μm, and washed to remove silt size particles. Magnetic minerals were removed using a Frantz Magnetic Barrier Laboratory Separator and the non-magnetic fraction was put in methylene iodide heavy liquids to obtain heavy mineral separates. Zircons were handpicked using a binocular microscope, and mounted in Struers Epofix resin and hardener and polished to expose their inner sections. Zircons were analysed using an ESI New Wave 193 nm laser system with a Trueline ablation cell connected by nylon tubing to an 8800 Agilent ICP-MS. Total He flow from the cell was 550 ml/min, and Ar carrier gas flow was 850 ml/min. Analyses were conducted in groups of 5, bracketed by 3 separate standard analyses. NIST 610 was used as a concentration standard, Temora 2 (TIMS $^{206}Pb/^{238}U$ age = 416.78 ± 0.33 Ma, Black et al., 2004) as the primary zircon standard, and Plešovice (ID-TIMS $^{206}Pb/^{238}U$ age = 337.13 ± 0.37 Ma, Sláma et al., 2008) as a monitor zircon standard.

The instrumentation setup including a laser fluence of 3.5 J/cm^2 and laser spot size with a 30 μm diameter, producing an average of 32 counts per second ^{207}Pb on a background of 0.83 counts per second for Temora 2, with U levels (413 ppm) common for zircon. The elemental Th/U average was 0.56. Eighteen isotopes overall were measured for all zircons (see Appendices 1 and 2) with a total dwell time of 0.40 s and half of it dedicated to U, Th and Pb isotopes. ^{235}U was calculated from ^{238}U assuming the ratio of 1:137.88. A fixed stoichiometric SiO_2 concentration for zircon (32.77 wt%) was employed as an internal standard to calculate trace element concentration. Iolite (Paton et al., 2011) and ISOPLOT (Ludwig, 2003) were used for data reduction with a common Pb calculation in Excel. Common Pb concentrations were calculated as a percentage of all ^{206}Pb using a ^{208}Pb basis, assuming common Pb composition (Cumming and Richards, 1975) and concordance of the Th–U and Pb–U systems. Because these calculated values for the concordant grains were <0.72%, no common Pb correction has been made. Errors in age calculations are reported at a 95% confidence interval using propagated errors (Paton et al., 2011).

4.2. Age of volcanic rocks from Boomi Creek

Our geochronological results are presented in Fig. 8 and Appendix 2. There were 28 successful zircon ablations of sample NT02, yielding 14 concordant ages (Fig. 8a). One zircon with anomalously old age of 391 Ma is interpreted as a xenocryst. This and three other analyses that displayed anomalous P and REE compositions were excluded from the weighted mean calculation, which yielded an age of 316.6 ± 3.3 Ma (n = 9, MSWD = 0.94, probability = 0.48; Fig. 8b) using propagated uncertainties (Fig. 8a). Their average U content was 374 ppm and elemental Th/U of 0.53.

Sample NT04 produced 25 successful zircon ablations, of which 14 ages were concordant (Fig. 8c). One zircon with anomalously old age of 355 Ma is interpreted as a xenocryst. This and one other analysis that displayed anomalous REE compositions were excluded from the weighted mean calculation, which yielded an age of 319.0 ± 2.9 Ma (n = 12, MSWD = 0.94, probability = 0.50; Fig. 8d) using propagated uncertainties. NT04 concordant grains had an average U content of 465 ppm and an elemental Th/U of 0.60.

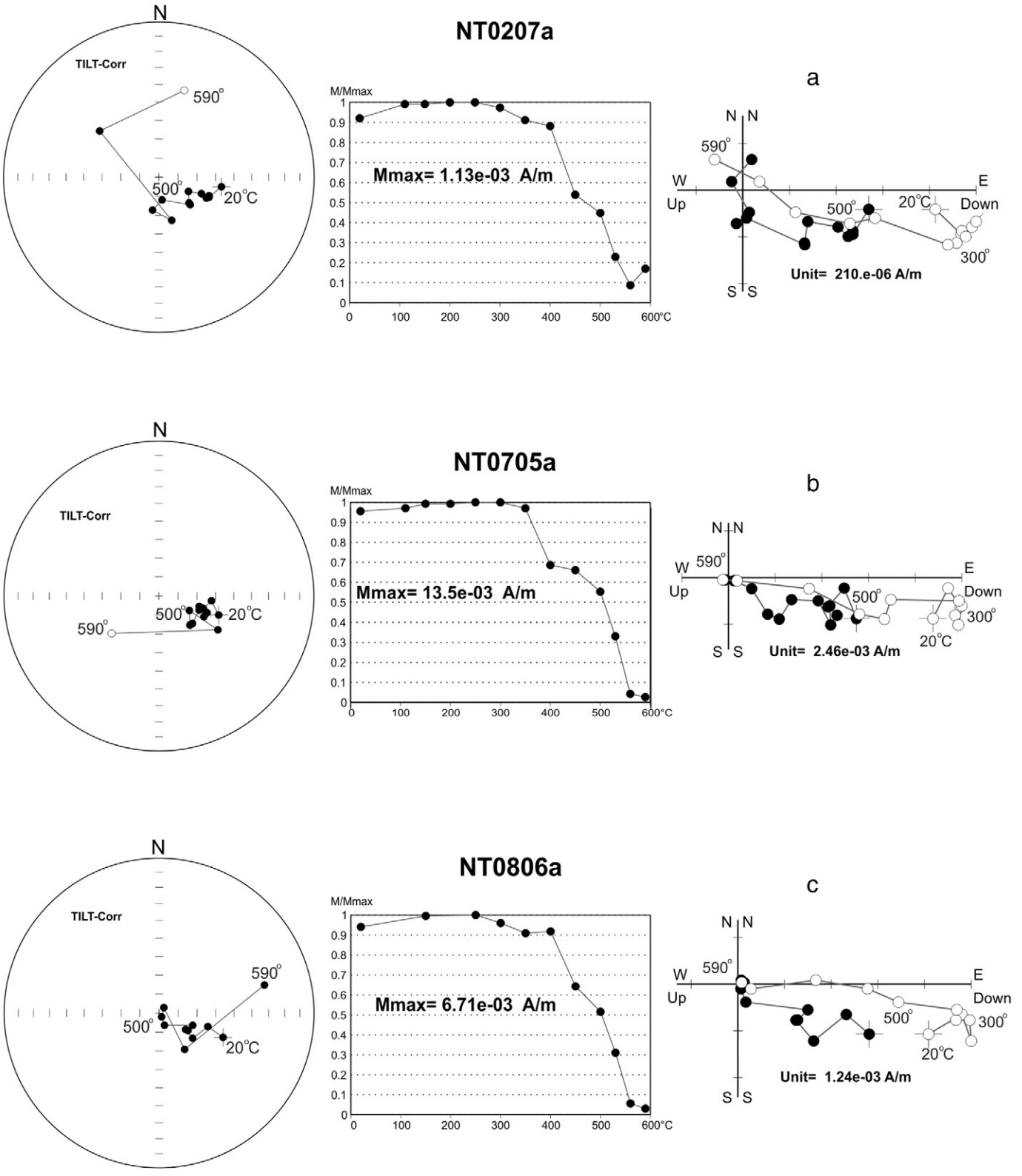


Fig. 6. Thermal demagnetizations of sedimentary rocks of the North Tamworth terrane. See notes to Fig. 2.

4.3. Analytical uncertainties

All analyses of the Plešovice monitor standard (see Appendices 1 and 2) were concordant in $^{207}Pb/^{235}U$–$^{206}Pb/^{238}U$ space within internal uncertainties, and yielded a weighted mean population $^{206}Pb/^{238}U$ age of 327.4 ± 0.23 Ma. Our measured age for the Plešovice is outside the accepted age of 337.13 ± 0.37 Ma (Sláma et al., 2008), perhaps due to a trace element dependent matrix effect. If the matrices of Plešovice and our Carboniferous zircons match, it is possible that the ages supplied here may be up to 3% too young. However, the Plešovice zircons have

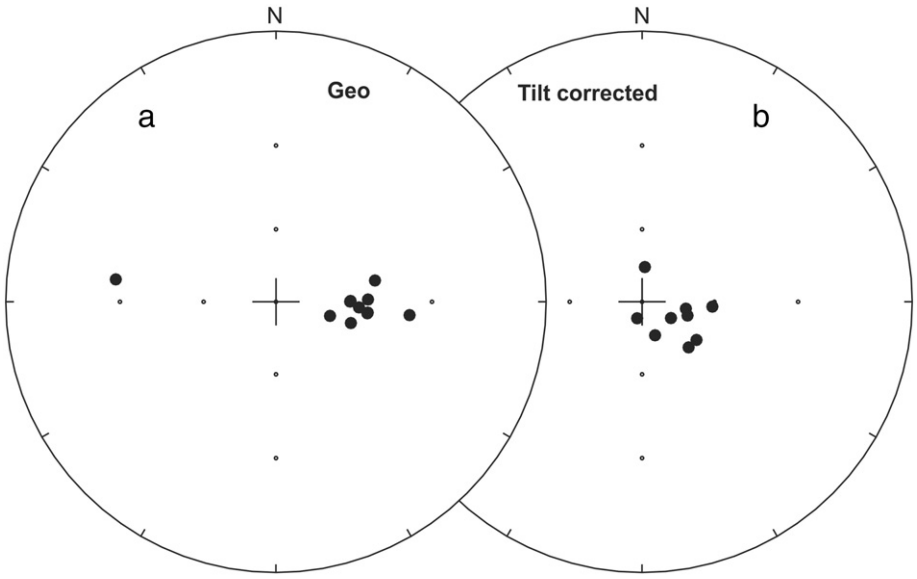


Fig. 7. (a–b) Stereoplots of the ChRM from the Rocky Creek samples, in situ and tilt corrected correspondingly.

significantly different trace element compositions relative to the Boomi Creek and Temora 2 specimens, including lower Th/U and Ce, as well as higher Ti and Lu, suggesting that the matrices are in fact quite different and that our ages are not affected.

5. Discussion

5.1. Northern Tamworth

New paleomagnetic and geochronological data from the northern Tamworth terrane enable us to refine earlier paleomagnetic studies from this terrane (Opdyke et al., 2000; Klootwijk, 2002, 2003). Cawood et al. (2011) compiled earlier data and combined them into two groups with loosely constrained ages of 322–302 Ma and 300–260 Ma. Although these data suggested relatively minor movements of the northern Tamworth terrane with respect to cratonic Australia (Gondwana), details of these movements were not clear (Cawood et al., 2011). Our new data from Boomi Creek, together with a significant progress in regional geochronology, provide an opportunity to verify some of these details with greater accuracy.

The new ages of 316.6 ± 3.3 Ma and 319.0 ± 2.9 Ma for the studied rocks rule out their belonging to the Caroda Formation, as it is mapped (Brown et al., 1992). The studied section in the Boomi Creek area is likely belonging to the Rocky Creek Conglomerate or Lark Hill Formation (Fig. 2). The similar trace element compositions and ages that are indistinguishable within error (Fig. 8a, b) suggest that for the age of the Boomi Creek paleopole, the data may be combined to give a better representative age of c. 318.0 ± 2.2 Ma (n = 21, MSWD = 0.95, probability = 0.52).

Using the updated stratigraphic scheme of the area (Fig. 2) we made a new compilation of previously published (Opdyke et al., 2000; Klootwijk, 2002, 2003) and new (Table 3) paleomagnetic data by combining them in narrower age groups (Table 4). The averaging has been done on the site level and we used only sites with three or more samples and $\alpha_{95} < 25°$.

In our paleogeographic reconstructions (Section 5.3) we use paleopoles from Table 1 of Cawood et al. (2011) for the Hastings, Myall, Gresford and Rouchel terranes, as well as a new paleopole (30.0°S, 153.2°E, A_{95} = 19.5°) from the 272 Ma Alum Mountain Volcanics for the Myall Block (Shaanan et al., 2015).

The new compilation (Table 4) shows four paleopoles for the northern Tamworth terrane with ages constrained for intervals of 10–15 m.y. This dataset is better than the previous compilation (Cawood et al., 2011), which only included two poles with less precise ages of 322–302 Ma and 300–260 Ma. Despite a large scatter (α_{95} = 18.8°), we calculated a mean c. 345–330 Ma combined pole for coeval Caroda and Merlewood formations (Fig. 2). We also calculated three separate poles for time intervals of 330–320 Ma (Clifden Formation), 320–305 Ma (Table 4: CRL) and 295–285 Ma (Lower Permian rocks studied by Klootwijk, 2003).

5.2. Emu Creek

The pre-folding Component H found in the c. 320–305 Ma rocks of the Emu Creek terrane (section 3.1; Table 2) is strikingly close to the mean direction of Currabubula, Rocky Creek and Lark Hill formations of the northern Tamworth terrane (Fig. 9a). This is surprising, as the two terranes are located on the opposite limbs of the Texas Orocline (Fig. 9b–c). This means that if the Component H is primary, then, according to our paleomagnetic data, the development of the Texas Orocline did not incorporate vertical-axis rotations. In other words, following the definition of Weil et al. (2004), the implication of Component H being primary is that the orogenic curvature in our study area is a primary arc and not an orocline. For a primary arc to be developed, a pre-existing curvature in the plate boundary must have existed, thus controlling the curved pattern of the Devonian–Carboniferous subduction system, as well as the emplacement of Early Permian granitoids that mimic the shape of the orogenic curvature (Rosenbaum et al., 2012). Curved subduction systems are indeed common (e.g., Frank, 1968), but their progressive development into a tightly curved plate boundary zone (e.g., Gibraltar Arc, western Mediterranean) inevitably involves oroclinal bending (Rosenbaum, 2014). It thus seems unlikely that the Texas Orocline has achieved its tight curvature without rotations.

An alternative explanation, recently proposed by Buckman et al. (2015), is that the map-view curvature in the southern NEO is an "apparent orocline", which resulted from the expression of a thrust carapace overlying exhumed accreted terranes. This explanation, however, is inconsistent with the recognition of steeply inclined structural features that mimic the curved shape of the Texas Orocline. These include curved steep structural fabrics (Li et al., 2012),

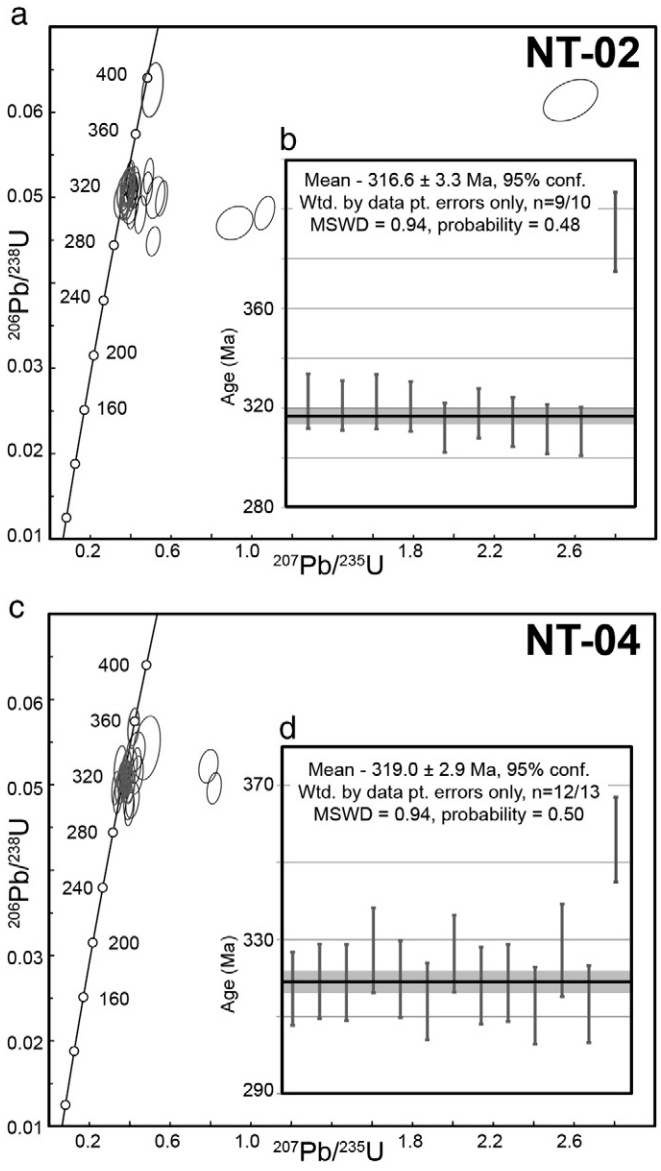


Fig. 8. Concordia diagrams (a, c) and weighted mean age plots (b, d) for samples NT02 and NT04, respectively. Concordant (Discordant) analyses are in red (grey), and inherited or anomalous zircons that were excluded from the calculation of the weighted mean age are in blue. Data-point error bars and ellipses are 2σ.

magnetic fabrics (Aubourg et al., 2004; Mochales et al., 2014) and the recognition of steeply dipping fault systems on both limbs of the orocline (Campbell et al., 2015; Babaahmadi and Rosenbaum, 2016). The ubiquitous recognition of steeply inclined structures throughout the Texas Orocline implies that the existence of a shallow thrust carapace is unlikely.

In light of the geological observations, we suggest that Component H is not primary even though it passes a fold test. A plausible explanation is provided below. Petrographic analysis of rocks from the Emu Creek and Paddys Flat formations corroborates field evidence that indicates that the entire succession has experienced one or more intense phases of silicification and sericitisation alteration (Hoy et al., 2014). With the exception of non-reactive quartz, the identification of clasts in the silicified rock is extremely difficult and most feldspar and lithic clasts have been replaced by clays. The alteration may be related to the emplacement and epithermal Au-mineralisation and alteration of the neighbouring volcanic and intrusive rocks at ~264 Ma (e.g. Drake Volcanics) or the intrusion of granitoids (e.g. Clarence River and Stanthorpe supersuites) at c. 250 Ma (Bryant et al., 1997; Cross and Blevin, 2010). The Emu Creek Block also hosted many small historical Au, Ag, As, Pb, Sb, and Cu polymetallic deposits as hydrothermal veins, stockworks, and wallrock disseminations (Brown et al., 2001), which may have acted as conduits for the hydrothermal fluids during this widespread alteration.

These events occurred during the Hunter Bowen Orogeny, which was deforming the NEO until ~230 Ma (Holcombe et al., 1997). Consequently, it is possible that the pre-folding remanence Component H has been acquired after oroclinal bending, but prior to folding. Lackie and Schmidt (1993) came to similar conclusions in their study of the Kiah Limestone of the northern Tamworth terrane. Details of this remagnetization are yet unclear, but we conclude that the tectonic history of Emu Creek terrane during the formation of the New England oroclines is still paleomagnetically unconstrained.

Another argument of the post-260 Ma age of Component H is the presence of both paleomagnetic polarities (Table 2, Fig. 5). Opdyke et al. (2000) put the base of the Kiaman Reverse Superchron within the Clifden Formation. All paleomagnetic studies of younger Currabubula, Rocky Creek and Lark Hill formations (Opdyke et al., 2000; Klootwijk, 2002, 2003; this study) show exclusively reverse (downward) remanence directions (Table 4), supporting the idea of their deposition during the Kiaman Reverse Superchron. If Component H is primary, it should be exclusively reverse too, as Paddys Flat and Emu Creek formations of the Emu Creek terrane are considered to be correlatives to the abovementioned formations of the northern Tamworth terrane (Fig. 2). The Kiaman Superchron ended at c. 265 Ma (e.g. McElhinny and McFadden, 2000), so the

Table 4
Paleomagnetic directions: formations of the North Tamworth terrane.

#	Formation	N	Decl.(°)	Incl. (°)	k	α_9 $(°)_5$	Plat (°S)	Plong. (°E)	Dp (°)	Dm (°)	Polarity	Reference
1	Caroda/Merlewood (c. 345–330 Ma)(**CM**)	16	166.0	31.6	4.8	18.8	71.2	284.8	11.8	21.1	9N7R	Opdyke et al., 2000; Klootwijk, 2002, 2003
2	Clifden (**CL**) (c. 330–320 Ma)	32	178.6	71.9	21.8	5.6	63.2	150.3	8.7	9.9	5N27R	Opdyke et al., 2000; Klootwijk, 2002, 2003;
3	Currabubula/ Rocky Creek/ Lark Hill (c. 320–305 Ma) (**CRL**)	112	214.5	75.9	30.2	2.5	50.1	150.3	4.2	4.6	0N112R	Opdyke et al., 2000; Klootwijk, 2002, 2003; this study
4	Lower Permian rocks (c. 295–285 Ma) (**LP**)	15	192.4	74.8	40.9	6.1	58.8	139.2	10.1	11.1	0N15R	Klootwijk, 2003

N = number of sites; Decl, Incl = site mean declination, inclination; Dp, Dm = the semi-axes of the cone of confidence about the pole at the 95% probability level. See also footnotes to Tables 1–3.

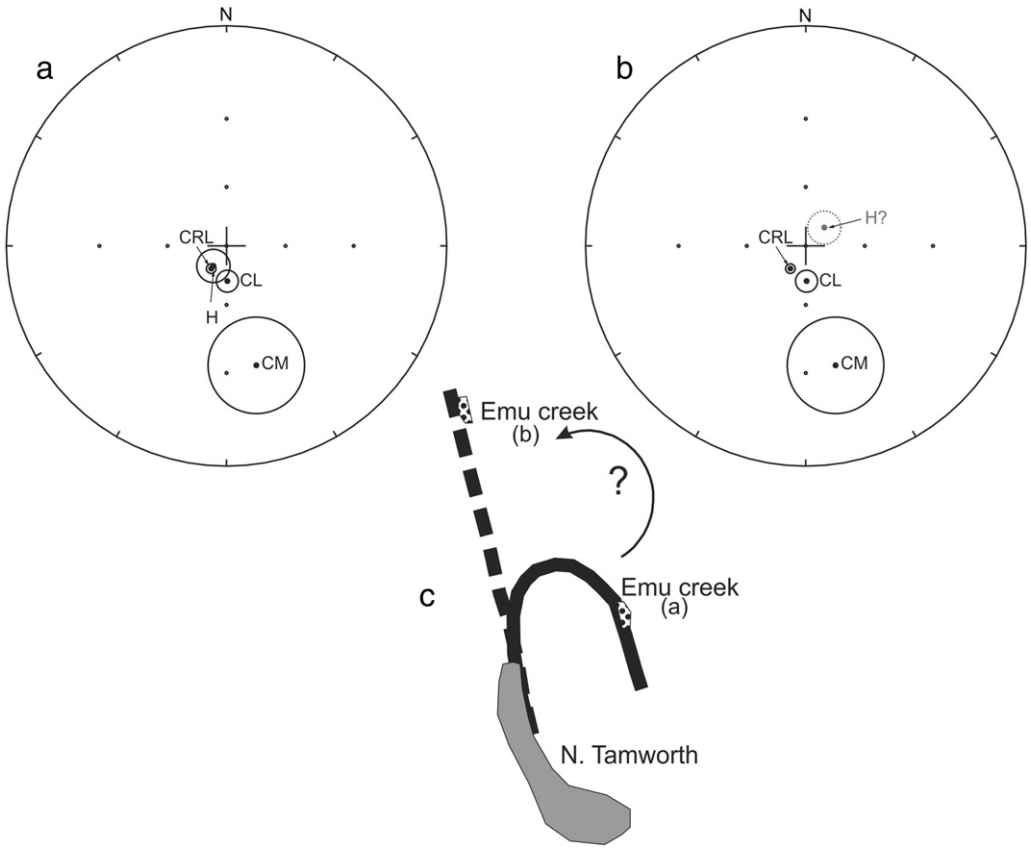


Fig. 9. (a) Mean paleomagnetic directions and their circles of confidence of c. 345–330 Ma Caroda/Merlewood (CM), c. 330–320 Ma Clifden (CL), c. 320–305 Ma Currabubula/Rocky Creek/Lark Hill formation (CRL) (Table 4), Component H in Emu Creek terrane (H) (Table 2). (b) The expected H direction (grey, dotted line) in suggestion that it is primary; the cartoon of oroclinal bending is shown in (c).

bipolar character of Component H is consistent with its post-265 Ma age.

5.3. Kinematic reconstruction

The new paleomagnetic data allow us to produce an improved kinematic reconstruction of the southern NEO, although we emphasise that such a reconstruction is non-unique. In a previous compilation of paleomagnetic data from the NEO (Cawood et al., 2011), a total of 19 reliable paleomagnetic poles for the time interval 340–260 Ma have been selected from the Global Paleomagnetic Database (Pisarevsky, 2005). Using these paleomagnetic poles, Cawood et al. (2011) have proposed a model, in which segments of the NEO were originally aligned on a quasi-linear >1000 km belt. This model suggested that the northern (in present coordinates) tip of this belt (North Queensland terrane) was fixed to the cratonic Australia, as supported by paleomagnetic data from this terrane (see figs. 7d and 7e in Cawood et al., 2011). The models showed that the original quasi-linear belt has been bent by sinistral transpression along the margin of cratonic Australia. The model of Cawood et al. (2011) is paleomagnetically permissive, but not unique, because of the scarcity of paleomagnetic data and their "diachronous" pattern (i.e., data from southern terranes of NEO were older than data from northern terranes). However, the analysis of Cawood et al. (2011) was sufficiently robust to rule out a previously suggested kinematic models involving oroclinal bending through dextral transpression (i.e., Offler and Foster, 2008).

The revised kinematic model for oroclinal bending in NEO (Fig. 10) is based on the compilation of Cawood et al. (2011), a new paleomagnetic pole recently obtained by Shaanan et al. (2015) and our new paleomagnetic data (Table 2; Table 4). The calculated Euler rotation parameters for the revised model are presented in Table 5. Mean poles for Gondwana, modified from McElhinny et al. (2003) are also listed in Cawood et al. (2011; Table 2). Our reconstruction, in contrast to Cawood et al. (2011), does not include a "Texas Block", because this area is not occupied with forearc basin terranes and is therefore not comparable to the other terranes shown in Fig. 10. An animated reconstruction is presented in the Supplementary Material (Appendix 1).

The main difference between our model and the model of Cawood et al. (2011) is the orientation of the northern Tamworth terrane at 340 Ma (Fig. 10a; see also Fig. 7a of Cawood et al., 2011). Cawood et al. (2011) did not use paleomagnetic data from Caroda and Merlewood formations (Opdyke et al., 2000; Klootwijk, 2002, 2003), and their 340 Ma reconstruction shows an elongated northern Tamworth terrane that continues along the western limb of the Texas Orocline. We calculated the mean Caroda/Merlewood (CM) pole by compiling 16 sites from Opdyke et al. (2000) and Klootwijk (2002, 2003). These directions are rather scattered (Table 4, entry 1), so this pole is less reliable than the Clifden (CL) and Currabubula/Rocky Creek (CRL) poles (Table 4, entries 2 and 3). However, to satisfy this pole, the orientation of the elongated northern Tamworth terrane should be nearly perpendicular to the suggested trend of the pre-orocline structure, whilst in the model of Cawood et al. (2011) this terrane is nearly parallel to that trend.

Consequently, there are two possibilities: (i) the pre-oroclinal configuration of forearc basins of the studies region was more complicated than suggested by Cawood et al. (2011), and/or (ii) the scattered paleomagnetic data from Caroda and Merlewood formations are not sufficiently robust. New paleomagnetic studies are required to resolve this problem.

The proposed 330–270 Ma evolution of the northern Tamworth terrane (Fig. 10b–e) is based on the newly compiled CL, CRL and (Lower

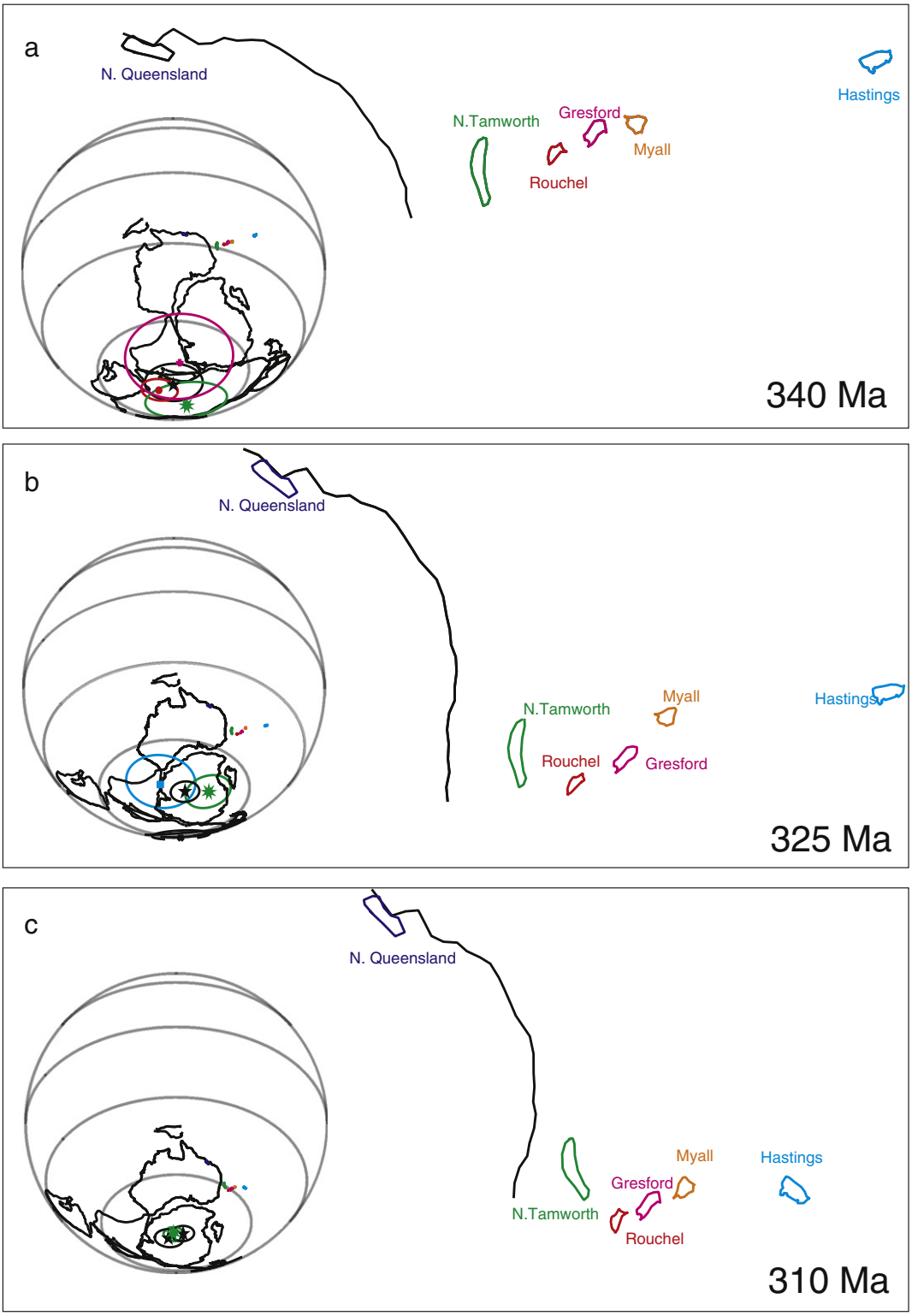


Fig. 10. The evolution of NEO between (a) 340 and (e) 270 Ma. Euler's poles of rotation of Gondwana, NEO terranes and paleomagnetic poles with their circles of confidence are listed in Table 5. Paleopoles are listed in Table 4 (northern Tamworth terrane), and in Table 1 (Hastings, Myall, Gresford and Rouchel terranes) and Table 2 (Gondwana) of Cawood et al. (2011). New c. 270 Ma paleopole for the Myall terrane (30.0°S, 153.2°E, A_{95} = 19.5°), recently reported by Shaanan et al. (2015) is shown in (e).

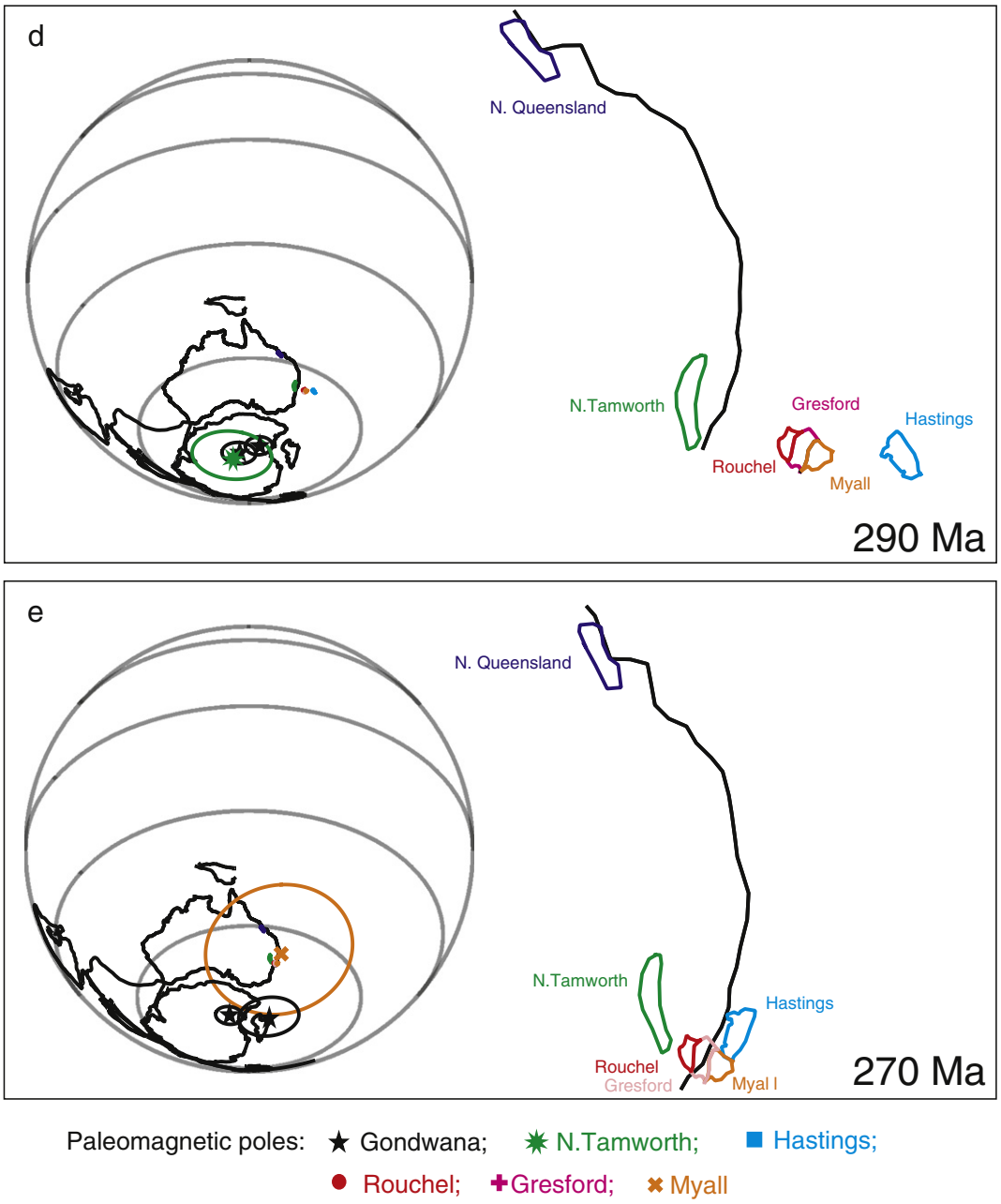


Fig. 10 (*continued*).

Permian) LP poles (Section 5.1; Table 4). There are a number of differences between this kinematic model and the previous model by Cawood et al. (2011). However, these differences are less significant in comparison with the abovementioned contrast in the 340 Ma reconstructions. The positioning of the Hastings Block in the 270 Ma reconstruction (Fig. 10e) is based on structural and chronological constraints that are described for the Nambucca Block by Shaanan et al. (2014).

6. Conclusions

Paleomagnetic data from Carboniferous sedimentary rocks of the Emu Creek terrane have revealed two secondary components of post-oroclinal age. In addition, new geochronological and paleomagnetic data from the Boomi Creek area of the northern Tamworth terrane have allowed us to improve the paleomagnetic dataset, which now allows us to present four paleopoles for the Visean–Artinskian rocks of the northern Tamworth terrane. The new poles allow us to further refine the kinematic model of the NEO, although further data are required in order to achieve a unique solution.

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.tecto.2016.01.029.

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Table 5
Euler rotation parameters for the terranes of New England Orogen.

Continent/terrane	Pole		Angle (°)	Rotated to
	(°N)	(°E)		
340 Ma				
Gondwana	38.63	−100.31	−122.33	Absolute framework
Hastings	65.02	−0.38	−84.66	Gondwana
Myall	37.94	−57.69	146.82	“
Gresford	25.70	−62.23	106.24	“
Rouchel	26.64	−61.77	112.77	“
N. Tamworth	15.77	−63.98	91.87	“
325 Ma				
Gondwana	44.70	−82.13	−105.11	Absolute framework
Hastings	65.94	−2.92	−83.69	Gondwana
Myall	29.35	−61.28	116.13	“
Gresford	20.21	−64.79	95.20	“
Rouchel	20.56	−64.68	99.71	“
N. Tamworth	3.55	−67.38	73.83	“
310 Ma				
Gondwana	47.99	−73.62	−99.71	Absolute framework
Hastings	68.60	53.30	−58.09	Gondwana
Myall	−0.64	−68.28	64.18	“
Gresford	11.00	−65.58	79.92	“
Rouchel	0.77	−67.81	68.52	“
N. Tamworth	−12.62	−69.17	58.28	“
290 Ma				
Gondwana	51.07	−70.40	−98.36	Absolute framework
Hastings	63.91	85.88	−52.24	Gondwana
Myall/Gresford/Rouchel	−6.31	−66.62	62.55	“
N. Tamworth	−1.17	−64.76	69.90	“
270 Ma				
Gondwana	54.55	−65.42	−92.54	Absolute framework
Hastings	59.11	99.09	−49.84	Gondwana
Myall/Gresford/Rouchel	−22.82	−66.75	53.73	“
N. Tamworth	−17.49	−65.93	56.75	“

Gondwanan configuration is after McElhinny et al. (2003). Cratonic Australia rotated to NW Africa at 28.1°S, 66.8°W, 52.1°. North Queensland is fixed to cratonic Australia. Positive angle of rotation = clockwise rotation.

reviewers for their useful comments. Paleogeographic reconstructions and animations are made with free GPLATES software (http://www.gplates.org/). This is the TIGeR publication #638. This is a contribution to IGCP 648.

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